

--1. What percentage of total revenue is generated by the top 20% of customers?—

	Top_20_percent_revenue
1	48.591317107788

--2. What is the month-over-month sales growth rate? Identify months with declining sales.--

	Month	Total_Sales	prev_month	growth_pct
1	2015-01-01	14205.7069849968	NULL	NULL
2	2015-02-01	4519.89195883274	14205.7069849968	-68.18
3	2015-03-01	55205.7974615097	4519.89195883274	1121.4
4	2015-04-01	27906.8548970222	55205.7974615097	-49.45
5	2015-05-01	23644.3028702736	27906.8548970222	-15.27
6	2015-06-01	34322.9354872704	23644.3028702736	45.16
7	2015-07-01	33781.5427848101	34322.9354872704	-1.58
8	2015-08-01	27117.5361847878	33781.5427848101	-19.73
9	2015-09-01	81623.5273054242	27117.5361847878	201
10	2015-10-01	31453.3927481174	81623.5273054242	-61.47
11	2015-11-01	77907.6605309248	31453.3927481174	147.69
12	2015-12-01	68167.0582851171	77907.6605309248	-12.5
13	2016-01-01	18066.957521677	68167.0582851171	-73.5
14	2016-02-01	11951.4109165668	18066.957521677	-33.85
15	2016-03-01	32339.3185715675	11951.4109165668	170.59
16	2016-04-01	34154.4684954882	32339.3185715675	5.61
17	2016-05-01	29959.5308282375	34154.4684954882	-12.28
18	2016-06-01	23599.373957932	29959.5308282375	-21.23
19	2016-07-01	28608.2589199543	23599.373957932	21.22
20	2016-08-01	36818.3421816826	28608.2589199543	28.7
21	2016-09-01	63133.6062759161	36818.3421816826	71.47
22	2016-10-01	31011.7373783588	63133.6062759161	-50.88
23	2016-11-01	70129.299262166	31011.7373783588	126.14
24	2016-12-01	74543.6015212536	70129.299262166	6.29

--3. Segment customers into High, Medium, and Low spenders based on total purchase value.--

	Customer_ID	Total_Spent	Segment
1	RB-19465	4715.46589171886	Low
2	JE-16165	2697.24798202515	Low
3	TS-21205	5371.09008026123	Medium
4	BW-11110	4750.35794353485	Low
5	SJ-20125	3949.65602874756	Low
6	CB-12025	5730.66897439957	Medium
7	GT-14755	226.436000823975	Low
8	MC-17845	3805.71398496628	Low
9	IL-15100	5040.7359316349	Medium
10	AG-10525	1455.03802061081	Low
11	FA-14230	2984.05400919914	Low
12	DP-13105	2215.00202941895	Low
13	DB-13660	1155.20196533203	Low
14	SN-20560	1860.41799783707	Low
15	KT-16480	4642.08843445778	Low
16	KD-16495	8181.25576925278	Medium
17	DI-13600	755.601972937584	Low
18	BV-11245	1523.27297735214	Low
19	SR-20425	3233.48111152649	Low
20	CS-11845	1278.95398974419	Low
21	VW-21775	6134.03801751137	Medium
22	RS-19765	4159.7699611187	Low
23	JR-16210	2201.68599796295	Low
24	MC-18100	2218.98004508018	Low

--4. What proportion of customers are repeat buyers vs one-time buyers?--

	Customer_type	Num_Customers
1	Repeat	780
2	One-Time	13

--5. What is the average delivery time, and which ship mode is the fastest and most consistent?—

--6. Do faster shipping methods (e.g., First Class) correlate with higher order values?--

Results			Messages		
	Ship_Mode	Avg_Delivery_Time			
1	Standard Class	5			
2	Second Class	3			
3	First Class	2			
4	Same Day	0			

	Ship_Mode	Avg_Order_Value			
1	Second Class	236.296148243796			
2	Same Day	232.74914181731			
3	First Class	230.228018670659			
4	Standard Cl...	227.836747536284			

--7. Which region generates the highest revenue, and which has the most consistent monthly sales?--

Results				Messages			
	Region	Total_Rev	Sales_Variability				
1	East	660589.353759289	621.617731739569				
2	South	389151.459338546	779.850557946155				
3	West	710219.683321834	524.240786978381				
4	Central	492646.912053883	636.040140764837				

--8. Which cities have high order volume but low average order value?--

Results		Messages	
	City	Total_Orders	Avg_Order_Value
1	New York City	891	283.347414640599
2	Los Angeles	728	238.214533997106
3	Philadelphia	532	204.589751533548
4	San Francisco	500	218.0822387743
5	Seattle	426	272.550051428623
6	Houston	374	171.005729114786
7	Chicago	308	155.260172018564
8	Columbus	221	174.943723216855
9	San Diego	170	279.53546463181
10	Springfield	161	259.800061529468
11	Dallas	156	129.025308993382
12	Jacksonville	125	357.705467556
13	Detroit	115	369.103860394851
14	Newark	92	309.217922026696
15	Jackson	82	304.437293788282
16	Richmond	81	218.072468186602
17	Columbia	81	312.139804327929
18	Aurora	68	171.418793578358
19	Phoenix	63	174.607252571318
20	Arlington	60	336.908870651325
21	San Antonio	59	370.229292245234
22	Miami	56	154.425472310611
23	Louisville	54	223.690482925486
24	Rochester	53	150.462829256957

--9. Identify products that generate high revenue but are purchased infrequently.--

	Product_Name	order_count	total_sales
1	Canon imageCLASS 2200 Advanced Copier	5	61599.822265625
2	Cisco TelePresence System EX90 Videoconferencing ...	1	22638.48046875
3	HON 5400 Series Task Chairs for Big and Tall	8	21870.5755004883
4	GBC Ibimaster 500 Manual ProClick Binding System	9	19024.5002441406
5	Hewlett Packard LaserJet 3310 Copier	8	18839.6859130859
6	HP Designjet T520 Inkjet Large Format Printer - 24" C...	3	18374.8952636719
7	GBC DocuBind P400 Electric Binding System	6	17965.0678710938
8	High Speed Automatic Electric Letter Opener	3	17030.3117675781
9	Lexmark MX611dhe Monochrome Laser Printer	4	16829.9011230469
10	Martin Yale Chadless Opener Electric Letter Opener	6	16656.1994018555
11	Ibico EPK-21 Electric Binding System	3	15875.9162597656
12	3D Systems Cube Printer, 2nd Generation, Magenta	2	14299.8896484375
13	Samsung Galaxy Mega 6.3	6	13943.6678466797
14	Apple iPhone 5	6	12996.6002197266
15	Global Troy Executive Leather Low-Back Tilter	9	12975.382019043
16	Riverside Palais Royal Lawyers Bookcase, Royale Ch...	4	11206.0656738281
17	Cubify CubeX 3D Printer Double Head Print	3	11099.9627685547
18	DMI Eclipse Executive Suite Bookcases	6	11046.6091918945
19	Logitech P710e Mobile Speakerphone	9	10196.6039428711
20	Chromcraft Bull-Nose Wood Oval Conference Tables ...	4	9587.05200195313
21	Plantronics Savi W720 Multi-Device Wireless Headset...	7	9367.29010009766
22	Bady BDG101FRU Card Printer	4	9279.88391113281

--10. Find the top-selling product in each month.--

	Month	Product_Name	Total_Sales	mk
1	2015-01-01	Global Deluxe High-Back Manager's Chair	2573.82006835938	1
2	2015-02-01	Hon 2111 Invitation Series Comer Table	1256.21997070313	1
3	2015-03-01	Cisco TelePresence System EX90 Videoconferencing ...	22638.48046875	1
4	2015-04-01	Martin Yale Chadless Opener Electric Letter Opener	4164.0498046875	1
5	2015-05-01	Fellowes PB300 Plastic Comb Binding Machine	2715.92993164063	1
6	2015-06-01	GBC DocuBind P400 Electric Binding System	3266.3759765625	1
7	2015-07-01	High Speed Automatic Electric Letter Opener	8187.64990234375	1
8	2015-08-01	Ibico Ibimaster 300 Manual Binding System	2060.74389648438	1
9	2015-09-01	Lexmark MX611dhe Monochrome Laser Printer	11219.9340820313	1
10	2015-10-01	Apple iPhone 5S	2735.95190429688	1
11	2015-11-01	Apple iPhone 5	4548.81005859375	1
12	2015-12-01	HP Designjet T520 Inkjet Large Format Printer - 24" C...	6999.9599609375	1
13	2016-01-01	Chromcraft Bull-Nose Wood Oval Conference Tables ...	4297.64404296875	1
14	2016-02-01	Fellowes PB500 Electric Punch Plastic Comb Binding ...	2541.97998046875	1
15	2016-03-01	Fellowes PB500 Electric Punch Plastic Comb Binding ...	6354.9501953125	1
16	2016-04-01	Fellowes PB500 Electric Punch Plastic Comb Binding ...	3812.96997070313	1
17	2016-05-01	Zebra GX420t Direct Thermal/Thermal Transfer Printer	2973.32006835938	1
18	2016-06-01	Fellowes PB500 Electric Punch Plastic Comb Binding ...	3050.3759765625	1
19	2016-07-01	Plantronics CS510 - Over-the-Head monaural Wireless...	2309.64990234375	1
20	2016-08-01	Cisco 9971 IP Video Phone Charcoal	3080	1
21	2016-09-01	Riverside Palais Royal Lawyers Bookcase, Royale Ch...	7312.13403320313	1
22	2016-10-01	HON 5400 Series Task Chairs for Big and Tall	2453.42993164063	1
23	2016-11-01	Zebra ZM400 Thermal Label Printer	6965.69970703125	1
24	2016-12-01	Canon PC1060 Personal Laser Copier	4899.93017578125	1

--11. Find the average number of days between consecutive orders for repeat customers.--

	Customer_ID	Avg_Days_between_Orders
1	AA-10315	118
2	AA-10375	95
3	AA-10480	97
4	AA-10645	72
5	AB-10015	199
6	AB-10060	45
7	AB-10105	56
8	AB-10150	123
9	AB-10165	85
10	AB-10255	83
11	AB-10600	155
12	AC-10420	258
13	AC-10450	143
14	AC-10615	48
15	AC-10660	178
16	AD-10180	100
17	AF-10870	47
18	AF-10885	152
19	AG-10270	80
20	AG-10300	306
21	AG-10330	63
22	AG-10390	176
23	AG-10495	77
24	AG-10525	49

--12. Calculate a 7-day rolling average of sales to identify trends and smooth fluctuations.--

	Order_Date	Total_Sales	Rolling_Avg_7_Days
1	2015-01-03	16.4479999542236	16.4479999542236
2	2015-01-04	288.059992790222	152.253996372223
3	2015-01-05	19.5359992980957	108.01466401418
4	2015-01-06	4407.1000828743	1182.78601872921
5	2015-01-07	87.1579971313477	963.660414409637
6	2015-01-09	40.5440006256104	809.807678778966
7	2015-01-10	54.8299987316132	701.953724486487
8	2015-01-11	9.9399995803833	701.024010147367
9	2015-01-13	3553.79497694969	1167.55757931301
10	2015-01-14	61.9599990844727	1173.61815071106
11	2015-01-15	149.949996948242	565.453852721623
12	2015-01-16	299.96399307251	595.854709284646
13	2015-01-18	64.8639984130859	599.328994682857
14	2015-01-19	378.59400177002	645.580995116915
15	2015-01-20	2673.86993694305	1026.14241474015
16	2015-01-23	40.0800018310547	524.183132580348
17	2015-01-26	1097.25002288818	672.08170740945
18	2015-01-27	426.669998168945	711.613136155265
19	2015-01-28	3.92799997329712	669.322279998234
20	2015-01-30	240.500002861023	694.413137776511
21	2015-01-31	290.665985107422	681.851992538997
22	2015-02-01	468.899993896484	366.856286389487
23	2015-02-02	211.646007537842	391.365715776171
24	2015-02-03	97.1119966506958	248.488854885101